Optimizing the Honolulu bike share System for First/Last Mile Infrastructure

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Project Outline

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1 – Abstract

We present this project to assist Biki, Honolulu's bike share system operated by Secure Bike Share, in addressing the first/last mile gaps in the city's public transportation network. Using community surveying, GIS mapping, and detailed data analysis, we developed strategies to strengthen the connection between Honolulu's bus system and the bike share network. We also created an improvement plan that identifies low-accessibility areas through suitability mapping, algorithmic station ranking, and targeted community input. Our recommendations include new way-finding signage at bus stops utilizing QR codes to monitor usage alongside nearby station activity, GIS-based expansion plans that account for distances to transit and existing bike routes, and data-driven tools for assessing network performance. Through these efforts, we aim to help Biki grow ridership and encourage bike sharing as a healthier, greener alternative to car travel in Honolulu.

2 – Introduction

Honolulu is the largest city in Hawai‘i, located in an ideal climate for biking and other forms of public transportation. Nevertheless, the city ranks among the worst for traffic congestion in the United States, with commuters losing an average of 64 hours per year to traffic (Schrank et al., 2019). As with much of the United States, Honolulu has become heavily dependent on private vehicles, contributing to urban sprawl, limiting parking availability, and increasing emissions of greenhouse gases. Despite its compact geography and generally favorable weather, the city's infrastructure has long prioritized cars over alternative modes of transportation. To combat these problems, there has been a recent push by residents and policymakers to shift Honolulu towards public transportation to improve the everyday transit experience. One of the key players in this process is Secure Bike Share, which operates Biki, Honolulu's bike share system. This project, in conjunction with Secure Bike Share, examines how Honolulu's bike share system, Biki, can better support public transit connectivity and accessibility through data-driven analysis of its bike network.

2.1 – Public Transit Context

The oldest form of public transportation in Honolulu is the bus system, aptly named "TheBus". Founded in 1971, TheBus has served as Honolulu's main form of public transportation for over fifty years. TheBus features 115 different routes and over 3000 bus stops across the entire island of O‘ahu, with most of these stops being located in the various urban areas of southern O‘ahu, as seen in Figure 1. We only treated bus stops located in and around the central areas where Biki operates. Notably, Biki and the city of Honolulu have begun developing plans to integrate the Holo Card, which enables easy payment for TheBus, with Biki's own network. This step would serve to integrate Biki with the city and to generally increase awareness of its brand.

Figure 1: Map of TheBus
(from thebus.org)

In addition to TheBus, the newest form of transportation in Honolulu is the Skyline rail system, which opened in 2023. Its goal is to provide long-distance public transportation across Honolulu. The rail, however, was still under construction at the time of our arrival and had yet to expand into downtown Honolulu and the Waikiki area, where we focused our project. The current extent of the system is shown in Figure 2.

Figure 2: *Honolulu Skyline Map*
(from honolulu.gov/dts/skyline/)

The final form of public transportation is the bike share system. Honolulu's bike share system, Biki, has over 100 stations, shown in Figure 3, and around 800 bikes.

Users can pay to take out a bike from any of their docking stations, and then return it to another docking station within the allotted time. They offer 3-gear bikes that are widely accommodating and maintained.

Figure 3: *Map of Biki Stations and Bike Lanes*
(graphic by the authors using data from Biki)

2.2 – Biki's History and Structure

The city of Honolulu started its large-scale bike share program in June 2017, when bike share Hawaiʻi, a non-profit designed to set the foundation of Honolulu's bike share system, and Secure Bike Share, which partnered with bike share Hawaiʻi in 2016, came together to form Biki. Secure Bike Share specializes in the management and deployment of large-scale bike share programs. Secure Bike Share themselves are operated by the award-winning Secure Parking Group. Bike share Hawaiʻi was granted two million dollars in 2015 from the state of Hawaiʻi to pay for the team's legal and accounting fees, market research, brand development, outreach, and site identification and planning.

While the city of Honolulu supported the structure and start-up of the company, Biki still faced financial challenges down the line. The COVID-19 pandemic caused significant hardships for Biki. Although the bike share system was marked as an essential business and therefore didn't shut down, it saw heavy losses in ridership due to lockdowns. Workers who would drive into town and then use bikes to travel short distances to meetings were now meeting virtually. Furthermore, students who often utilize the bike share system were no longer attending classes at their universities. These losses reflected in a 50% decrease in bike trips during the pandemic. In order to deal with these losses, seven stations had to be decommissioned, and services like call center operation hours had to be reduced.

On July 1st, 2024, bike share Hawaiʻi transferred Biki's oversight to the City and County of Honolulu's Department of Transportation. Up until that point, the non-profit, bike share Hawaiʻi, helped to establish the bike share system and implement up to 1,288 bikes, 136 self-service stations, and 2,500 docked bicycle stalls, making it the sixth most used bike share system in the United States. Although oversight has been transferred to the Department of Transportation, Biki's partner, Secure Bike Share, still operates and maintains Biki's bike system. This transition in oversight is expected to create new opportunities for expansion, as the Department of Transportation can authorize the placement of Biki stations on city property (Yamanaka, n.d.).

Unlike other bike share systems, in order to remain self-sufficient, Biki uses the revenue from the passes it sells rather than requiring subsidies from the city. Biki offers both local commuter passes and short-term visitor options. As for Biki's ridership, the most recent information is taken from Biki's 2023 rider data. There were a total of 797,953 rides, 62% of which were taken by residents of Honolulu. The system had a total of 75,790 riders, 17,079 active members, 10,962 of which were new members. Users of Biki have reported many benefits to using the bike share system, the consensus being that the system helped users save money and exercise more. In 2023, Biki was able to reach some impressive milestones as well, such as 1.6 million miles traveled, 1.9 million lbs of CO2 avoided, and 65 million calories burned (Biki, 2024).

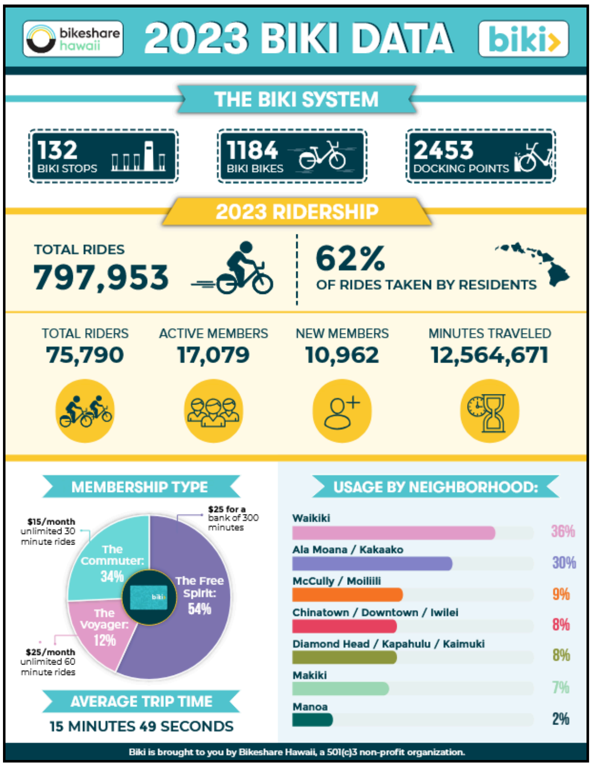

Figure 4: *Biki data info graphic* (graphic by Biki)

2.3 – Station Deployment

Many of Biki's stations were originally installed in 2016 during the program's launch. At the time, the priority was rapid deployment to establish the bike share network, which meant limited opportunity for in-depth infrastructure analysis. Since then, station adjustments have been minimal and largely reactive, typically driven by permit expirations, construction, or other site-specific needs rather than strategic planning. The stations can be physically moved within a day, but other obstacles, like hiring a forklift and getting the necessary paperwork, can draw out the moving process for longer.

2.4 – Equity and Tourism

Bike sharing systems have large implications for the equity and accessibility of transportation, eliminating the "barriers to bike ownership such as the lack of safe parking spaces for bikes in cities, vandalism, and theft of bicycles, and the inconvenience and cost of owning and maintaining a bicycle" (Kabra et al., 2020). The social mobility provided by these lower-investment bike-transit systems has a large impact on low-income populations, providing methods for traveling to job sites that other forms of transport may not cover fully (Kosmidis et al., 2025).

It is imperative that bike sharing opportunities are offered equally to both low and high-income households. Drawing from recent studies in Sacramento, we conclude that low-income users, especially those living in zero-car households, are less likely to start using bike sharing networks, but utilize the service with higher frequency once they start using it (Mohiuddin et al., 2023). The ease of affordable transportation offered by bike sharing is appealing to disadvantaged communities and households without cars, and correlates with higher frequency of use. It is also important to note that "the disproportionate use of bike share services by higher-income individuals is partly due to the disproportionate siting of bike share stations in wealthier neighborhoods" (Mohiuddin et al., 2023). Clearly, a big problem in many bike sharing networks remains the introduction of bike stations to low-income areas.

One of the ways Biki helps to alleviate this issue is through their program 'Everybody Rides'. This program is available to Kamaʻāina who are currently receiving some sort of public assistance, such as SNAP, TANF, ONF, GA, Child Care Subsidies, or AABD. It provides unlimited 30-minute rides per month at only $10/ month, a 60% discount from the usual cost ("Everybody Rides," n.d.). This service currently averages around 400 trips a month with fewer than ten unique users.

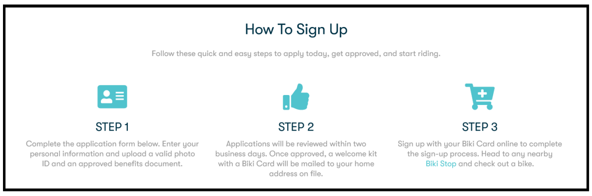

Figure 5: A screenshot from Biki's website detailing the process to apply to its "Everybody Rides" program (graphic by Biki)

3 – Methods

This project, in partnership with Secure Bike Share, aimed to close the first/last mile gaps in the Honolulu bike share system. By evaluating how well existing bike stations are integrated with Honolulu's bike lanes and transit systems, we recommended improvements based on findings from user surveying, journey mapping, and geographical information systems (GIS). By integrating bike stations with infrastructure, we aim to improve the transit experience of Biki users to incentivize bike share use. As our final deliverable, we produced a prioritized improvement plan for bike station integration with supporting GIS visuals, alongside user feedback and a proposal for enhanced bike station way-finding signage from bus stations.

3.1 – Objective 1: Prioritized Improvement Plan

In order to produce a prioritized improvement plan, we needed to determine: (1) suitable locations for new/moved bike stations to be implemented. (2) bike stations of low importance that can safely be moved to these more suitable locations. To accomplish this, we utilized ArcGIS mapping software and data analysis to create station rankings.

3.1.1 – Suitability Maps

A suitability map is a tool in ArcGIS Pro that enables the user to create a map with an overlaid gradient dependent on different geographical factors. These factors can range from distance to certain points or terrain elevation. These data sources are rasterized and weighted in order to produce a final result, which ideally has distinct zones that can be further investigated. Our suitability maps used data from Biki, TheBus, HonoluluGIS, and the Honolulu census.

3.1.2 – Ranking Bike Stations

Our second objective was to identify weakly integrated bike *stations*. Due to Biki's budgetary constraints, we wanted to offer a budget-conservative way to add stations to highly suitable locations. By quantifying which stations are least important to the bike sharing network, we provided Biki a tentative list of possible stations to relocate to areas with high suitability, while maintaining a minimal potential negative impact.

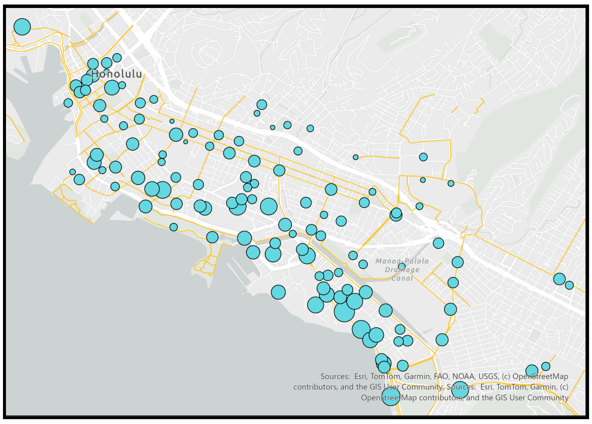

Figure 6: A GIS map depicting the ranking of individual bike stations. Larger dots represent a higher importance (See Appendix A for more details on the ranking function)
(graphic by the authors)

The ranking algorithm we implemented took into account both the ridership of a given station and its distance to other stations to get a measure of the impact that removing the station would have on the total system. If a station has a high rank, implying good ridership and vicinity to other stations, then the station's removal would have a higher negative impact on the network when compared to a low-ranking station (See Appendix A for implementation details).

3.2 – Objective 2: Surveying

In addition to data analysis, our team decided to survey Biki users. (See Appendix E for a flowchart of our survey's structure)

In compliance with WPI's Institutional Review Board (IRB 26-0117), all participants read an informed consent preamble before completing the survey.

On November 13th, Secure Bike Share sent out an online survey to over 20,000 people on our behalf through their mailing list. Recipients were split into six different geographical zones, and all responders who submitted a form were offered a $5.00 discount code in order to incentivize responses. The survey's primary purpose was to gauge the public's opinion on their experience with bike sharing in Honolulu, while also gathering data that could be used in our improvement plan. The survey was specifically designed to gather information that directly correlated to findings from our data analysis. The combination of analysis with public census enabled us to ensure our improvement plan was aligned with public interest.

3.2.1 – Survey Design

Our top priority for our survey was to collect data that we could conditionally differentiate based on certain demographics in order to draw strong conclusions. We were acutely aware that out of the users who would fill out our survey, the demographic would be greatly skewed towards dedicated bike share users. By designing our survey in a way that would allow us to narrow down responses based on demographics, we could avoid broad generalizations and ensure that each sector got equal representation.

One key example of this conditioning was our differentiation of responses between residents of Hawaiʻi and visitors to the islands. Like many other Hawaiian services, Biki places a strong emphasis on its local (Kamaʻaina) users. This was especially significant later on in our analysis, as we could empirically conclude that long-term, local users necessarily form the backbone of successful bike share systems (See Appendix F for a novel approach for determining this). However, it is also important to consider the experiences of visitors to the islands, as tourists make up a large portion of Biki's revenue. Ultimately, both locals and visitors bring helpful perspectives when it comes to our project, as first-time tourist users provide information about first impressions, while local users provide information about long-term issues.

Finally, an aspect of our survey that we wanted to emphasize was the ability for people to give as much input as they wanted for certain questions, particularly ones about bike share usage and perceived barriers. To accomplish this, we left many questions open-ended by allowing an "other" option for multiple choice responses, as well as adding a final free-response question asking about the user's impressions of bike sharing in Honolulu. In this manner, responders who wanted to efficiently fill out the form could do so, while those who wanted to give it more thought could also express their concerns in more detail.

It is important to note some of the limitations of this survey. Even though we collected a large sample of data, it all came from existing users of Biki. This can be useful as they are most likely to have useful feedback, but it is also a smaller demographic in general, with a predisposition to bike riding.

3.3 – Objective 3: Way-finding Bus Signage

Our final objective was to improve Biki's way-finding methods with new signage. Our signage aimed to increase the awareness and accessibility of bike sharing options. Due to our focus on closing the first/ last mile gaps, we decided to pilot our signage plan at bus stops to help connect them with Biki's network.

3.3.1 – Design

For our signage design, we wanted to produce a simple, easy-to-read sign that would direct users at bus stops in the direction of the closest bike station. We utilized a free open source image editing program, *Pixlr*, to create our mockups for the signage. In addition, we had to follow Biki's guidelines to ensure the signage fit with the grander Biki brand. Some of the guidelines we had to fulfill were displaying a graphic on a bike, adding a picture of Biki's logo, and having a scannable QR code that would redirect to Biki's interactive online station map.

3.3.2 – Site Selection

Our goal with the signage was to measure how well bus stops connect with bike stations, and how distance affected this connection. To this end, the signage acted both as our way of further integrating the bus and bike share networks, as well as a way of a way to collect data on this connection. We developed a method for gauging the success of the signage. We identified the following test categories

1. Bus stops with bike stations within zero to one-sixteenth of a mile
2. Bus stops with bike stations within one-fourth to one-half of a mile

From these two categories, we ranked the suitability of each bus stop based on a high number of daily riders, a low deviation of riders from season to season, and a proximity to highly central bike stations (See Appendix B for more details). We then chose five stops from both categories to be tested, and five stops from each category to act as controls. We aimed to measure both the number of QR code scans and the number of bike rides originating from stations close to the bus stop within 5 minutes of the bus arriving. In an ideal scenario, we would have measured the number of users who both scanned the code *and* proceeded to use a bike station, but this information was unavailable to us.

Tracking the amount of QR code scans at a bus stop allowed us to approximate the number of people who are *interested* in bike sharing, yet are *unaware* of the bike sharing options in the area of the bus stop. The amount of bike ridership following a bus stop's arrival represents the *willingness* of bus users to use bike sharing options following the bus stop.

While awareness is generally easy to measure using the amount of QR code scans, measuring how signage and distance affect *willingness* is more difficult. Having selected bus stops with high ridership and low variability, data on the amount of bus-to-bike ridership could be compared between test and control stops belonging to the same category. Seeing a higher overall ridership in the test group corresponding to high QR code scans would indicate that the signage was successful. Additionally, comparing how the success of signage between both distance categories would provide information about the necessity of nearby stations to bus stops for first/last mile connections.

4 – Results

In the following sections, we report on the results for our objectives. We spare many details in the section. Address our improvement plan and modeling information for more details.

4.1 – Objective 1: Prioritized Improvement Plan

4.1.1 – GIS maps

The first graphic we created was a simple suitability model that indicated promising areas for new bike stations. Figure 7 utilizes a green-red scale for suitability, green implying a more suitable location. The total score was proportional to:

- Distance from Biki Stations
- LODES Traffic

And inversely proportional to:

- Distance to Bike Lanes
- Distance to Bus Stops

These factors create a map that highlights locations with potential for expansion while still being connected to the transit ecosystem. The areas highlighted here are primarily in the north-western part of our study area, around Chinatown.

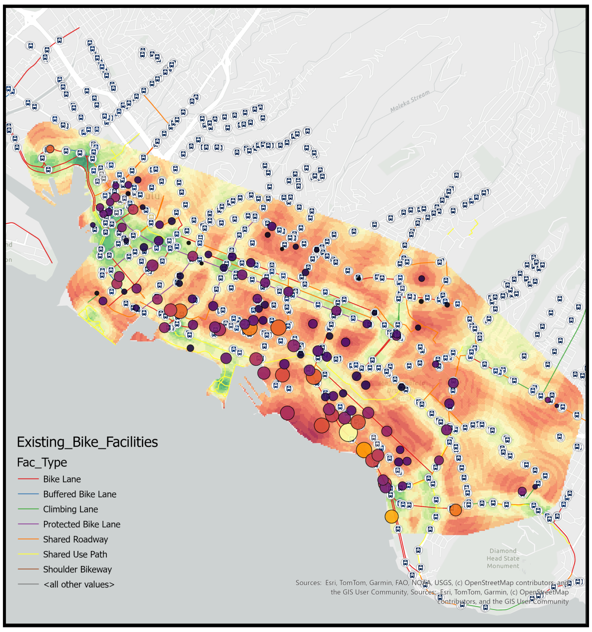

Figure 7: *Suitability Map 1*
(graphic by the authors)

The second map we created was similar to the first; however, we wanted to highlight any gaps in bus coverage that may exist. Figure 8 utilizes a green-red scale for suitability, green implying a more suitable location. This time, the score was proportional to:

- Distance to Bike Lanes
- Distance from Biki Stations
- LODES Traffic

But inversely proportional to

- Distance to Bus Stops

This creates a map that finds any potential areas that aren't covered by the bus, but have high demand and good bike infrastructure.

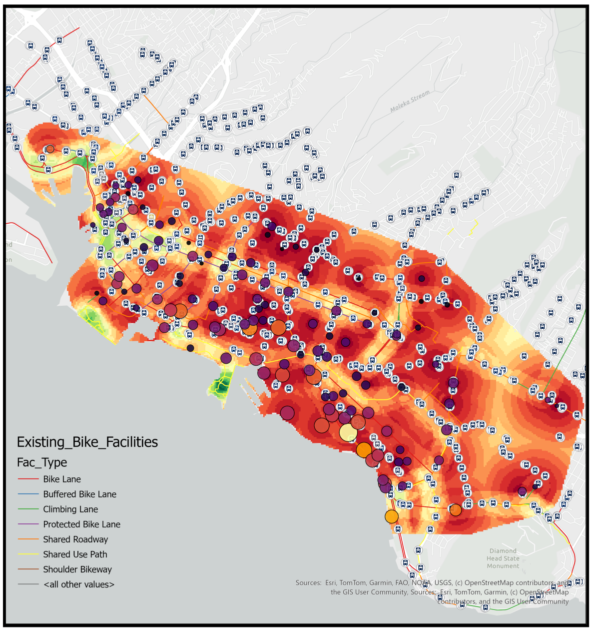

Figure 8: *Suitability Map 2*
(graphic by the authors)

Figure 9 demonstrates the connections with bike lanes. We gave bike lanes a ranking based on their type, i.e., protected lanes vs. mixed roadways.

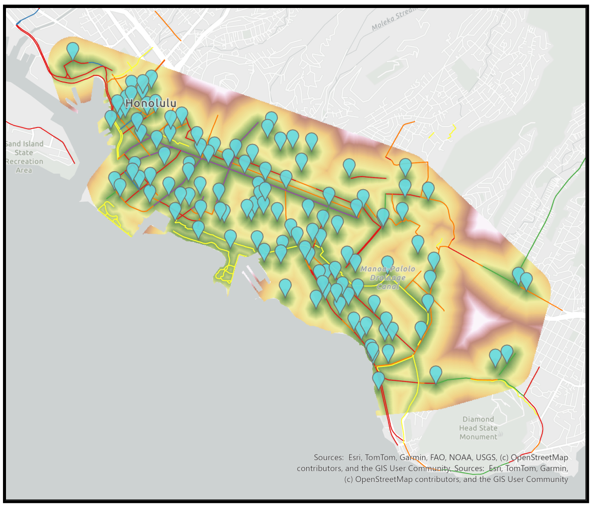

Figure 9: *Map of Bike Lane Connectivity*
(graphic by the authors)

We then used this information to create a cost-distance analysis. This analysis highlights any areas in green that are along easy-to-traverse bike routes and are close to bike stations. The resulting highlighted regions were a few locations in eastern Honolulu near Diamond Head, and a few other bike lanes that currently don't have a Biki station.

By combining these GIS maps with our ranking algorithm, we were able to suggest to Biki areas for future expansion, as well as stations that could be relocated to these higher demand areas

4.2 – Objective 2: User Feedback Survey

In addition to our GIS mapping, our survey showed us conclusive results on people's perceived barriers towards bike sharing in Honolulu on an individual level. As seen in Figure 10, 143 out of 233, or ~61% of responses, believed that a lack of bike-friendly infrastructure was a significant barrier for bike sharing in Honolulu.

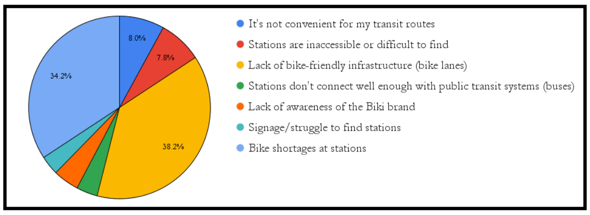

Figure 10: Barriers to Use
(graphic by the authors using data from the Biki User Survey)

Additionally, ~54% of responses believed that a lack of bikes at stations was an issue, which is to be expected due to the budgetary constraints highlighted before.

4.2.1 – Survey Usage

We collected all of our survey zone data into a master spreadsheet for analysis. There, we produced a significant number of graphs and supporting visuals for usage in our improvement plan. Within this spreadsheet, we also created filters for our results based on the different questions in order to ensure that we were using the most relevant data to draw conclusions. We knew that our survey would primarily draw responses from dedicated bike share users. By conditioning our results based on whether or not the responder used bikes as a primary form of transportation, for example, we could equally represent all demographics of responders. This conditional analysis enabled us to consider both big picture data as well as specific cases.

Our most powerful deliverable using this method compared the results from people who preferred to use the bus versus people who preferred to use bike systems, as seen in Figure 11. Data pointed towards frequent bike share users also being more frequent bus users, while frequent bus users did not use the bike share quite as much, helping us to verify the necessity of increasing wayfinding signage to help bus users close the last mile of their transportation with Biki.

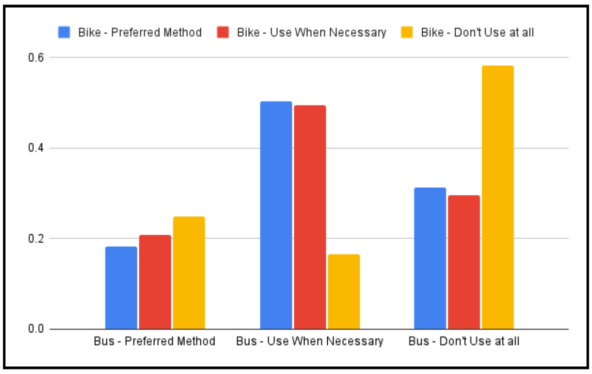

Figure 11: Biki and TheBus users Preferred Transit Methods
(graphic by the authors using data from Biki User Survey)

Figure 11 shows that bike share users are also bus users. Clearly, responders who said they were unlikely to use Biki were also unlikely to use TheBus.

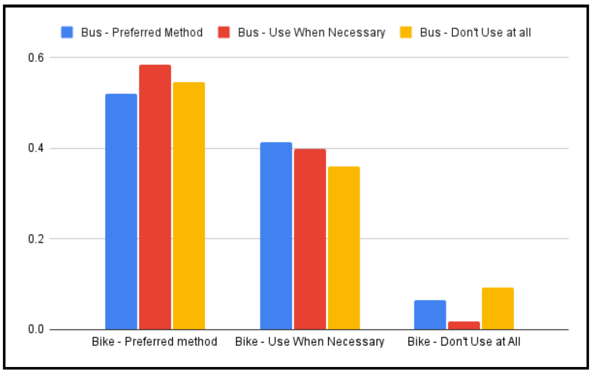

Figure 12: Biki and TheBus users Preferred Transit Methods
(graphic by the authors using data from Biki User Survey)

On the other hand, bus users were not necessarily bike users, as seen in Figure 12. Conditioned on bus usage, the data showed that responders who were more likely to use the bus were no more likely to use Biki.

4.3 – Objective 3: Way-finding Bus Signage

When beginning the design process for signage, there were three characteristics that needed to be displayed. These being that the signage had to draw the viewer's attention to a QR code leading to Biki's interactive map, that the design followed the rules of Biki's official branding guide, and that a bike was easily identifiable within the design.

Many variations of the signage were created, and our current designs are vastly different than our original iterations, which had been created for the sole purpose of improving bike station visibility, rather than increasing awareness about bike station locations at bus stops (see Appendix C). As displayed in Figure 13, there are directional variations of bike signage for both left and right for signs that are to be attached to the front of a rest stop, facing towards the street, indicating which way to the closest station. While discussing plans for implementing signage at different bus stop locations with employees at TheBus, we were made aware of minimum height regulations for signage to be attached to sign poles. The bottom of the sign needs to be at least 7 feet from ground level. For the last sign variant in Figure 13, the formatting of the signage is horizontal to ensure that we maximize the visibility of the sign while also maximizing the height of the sign. Also, since bus stop signage is placed perpendicular to roads, the arrow on this variation points upward, meaning the station is in front of the sign, rather than to either side.

Figure 13: Vertical bus stop sign (graphics by the authors)

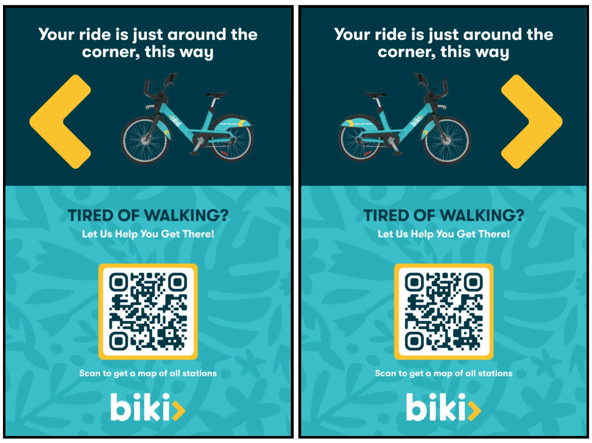

Figure 14: Biki signs detailing the direction to the selected connecting bike station and a QR code directing to the online Biki station map
(graphics by the authors)

As of the end of this project, approval for signage implementation from TheBus is pending. The pilot phase will collect data for 3 months by our sponsor after our project's end date.

5 – Discussion

Bike Share systems have lots of potential to improve local transit ecosystems by filling first/last mile gaps and reducing reliance on cars. Prior research shows that bike share ridership relies upon access to safe, reliable, and consistent bike lanes (NATCO, 2016). These studies closely relate to what we observed taking place in Honolulu. Our findings have reinforced this knowledge while adding insights that relate to Biki's system.

After leaving Honolulu, Biki will continue our work, specifically in regards to the testing of signage. While we oversaw the design and execution of the signs, we did not have the opportunity to observe the results over a long period of time. Since the time frame for when approval of signage placement is out of our control, this limited our project from being able to analyze the data and produce results demonstrating the signage's effectiveness. Because of this, we left Biki with the tools and methodologies to use the resulting data. They plan to run trials over a 3-9 month period, determining whether or not bus stop signage is something they would like to pursue further in the future.

While the focus of our project was on the connections of the bike share system with local infrastructure, we also observed other issues that need to be addressed through community input. From our survey, we found that a large number of reported barriers to using bike share were the lack of bikes. Users reported frequently going to take out a bike only to find empty or non-operational stations. If we assume that the optimal number of bikes for a system is half the number of docks, then Biki's network should employ roughly 1200 bikes. However, they currently only have around 750 bikes in service. The lack of bikes at certain stations is also affected by the limited amount of bike rebalancers due to Biki's budgetary constraints. Currently, only one rebalancer truck is in operation, rather than the two that were originally employed.

There were also plenty of users who were happy with the system, but simply wanted it to expand and add more options. Many wished for Biki to expand into more neighborhoods and expand further into certain areas of Honolulu. There was also a large desire for additional options such as e-bikes, surfboard holders, bike phone holders, and other bike accessories. While these wants are important to acknowledge, we recommend that Biki focus its time and money on the deployment of more bikes into its system.

As displayed in the survey results, the largest barrier for users was the lack of bike-friendly infrastructure. During our stay in Hawaiʻi, we noticed on group rides that many of these infrastructure issues were due to bike lanes being unguarded and because of the major gaps between bike lanes. While this project focused on closing the first/last mile gap, it appears that the average user finds the issue of rideability in relation to bike supply and infrastructure more pressing than connection with alternate forms of transportation.

While results from our survey analysis showed that there needs to be a greater focus on bike supply and infrastructure, raising awareness of Biki's brand may still be an effective way to increase ridership. Currently, signage is pending approval; however, if results show that many people scan our QR codes, then that means there is an untapped user base that is interested in bike share but is unaware of the service or how to find it. If this is true, Biki could expand even further by raising awareness of the Biki brand by not only placing signage at bus stations to help with wayfinding, but also by including advertisements within the buses themselves. This may further help with wayfinding for users since it gives the user time to plan a route while riding the bus, making them more likely to engage with the bike system when departing.

Since this project focused mostly on closing gaps between stations and bus stops, rather than on bike rebalancing and infrastructure, a similar approach could be taken in other cities with bike share systems to achieve greater results, assuming that these locations have good infrastructure and bike supply, but still need to optimize their station placement. Working with mapping software such as *ArcGIS* helps to generate graphical representations of rider data to better expose these system gaps by analyzing traffic through stations, bike station rankings, bike route locations, etc. Furthermore, this project also helps to outline effective ways of determining the best candidates for signage deployment at bus stops. This may serve to be helpful for smaller bike share companies that may have limited amounts of funding to fabricate and install signage. Ensuring that the best locations are chosen for the initial deployment of signage.

We must also acknowledge several limitations within our work. A primary constraint was the timeline regarding the approval process for public infrastructure. Because approval from TheBus is pending, we could not execute the data collection phase of the signage pilot prior to the end of the term. Consequently, the quantitative efficacy of the way-finding signage in bridging the First/Last-mile gap remains theoretical until the conclusion of the proposed six-month trial. Additionally, despite our efforts to filter results based on demographics, our survey data contains inherent self-selection bias. Because the survey was distributed via Biki's existing mailing list, the respondents were predominantly active users, meaning the specific barriers faced by non-users or residents who have abandoned the system may be underrepresented. Finally, our station ranking algorithm (Appendix A) relies on heuristic weighting parameters (α, β, γ). While we tuned these variables to reflect Honolulu's specific traffic patterns, they remain mathematical approximations that assume distance and ridership are the sole primary drivers of station importance, potentially overlooking complex external factors like elevation changes or specific tourist behaviors.

In conclusion, this project has demonstrated that using signage that targets potential Biki users can help with improving way-finding and increasing awareness of the Biki brand, the broader success of the system mainly depends on addressing the gaps in infrastructure for bikes and on the deeper operational challenges. By providing Biki with a methodology for long-term signage deployment and identifying key barriers such as limited bike availability, insufficient rebalancing, and breakage in bike routes, we lay the foundation for the best plan of action to improve on Biki's system and its integration within Honolulu. The survey results highlighted both the community's enthusiasm for expanding and creating a more accessible bike share network, as well as the critical need for investments in supply and safety to support the growth and maintenance of the current system. The methodology of this project can be utilized in other cities that are trying to strengthen multi-modal transportation through data-driven signage placement, as well as to improve station placement through a ranking system. In its entirety, the results of this project help to create thoughtful system planning and placement of stations, increase the overall awareness of the brand via signage, and suggest future improvements for Biki's bike share system, based on popular demand from surveys.

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